



B.TECH 1ST YEAR · 1ST SEMESTER · LAST-DAY REVISION

# Engineering Chemistry Formula Master Sheet

Unit I – Structure & Bonding Models + Electrochemistry



Quantum Theory



Wave Mechanics



Molecular Orbitals



Electrochemistry



25 Flashcards



Exam Tips

$$E = h\nu \quad | \quad \lambda = h/mv$$

$$\text{Bond Order} = (N_b - N_a) / 2$$

$$E = E^\circ - (0.0591/n) \log Q$$

$$\Delta H_\psi = E_\psi \quad | \quad \psi^2 = \text{Probability Density}$$



## 1. Planck's Quantum Theory

### CORE FORMULA

$$E = h\nu$$

### ALTERNATIVE FORM

$$E = hc / \lambda$$

### Key Assumptions

- Energy is emitted/absorbed in discrete packets (quanta)
- Each quantum has energy  $E = h\nu$
- Energy is NOT continuous — it's quantized

| Symbol    | Quantity                 | Value / Unit                                   |
|-----------|--------------------------|------------------------------------------------|
| $h$       | Planck's Constant        | $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ |
| $\nu$     | Frequency                | Hz ( $\text{s}^{-1}$ )                         |
| $c$       | Speed of light           | $3 \times 10^8 \text{ m/s}$                    |
| $\lambda$ | Wavelength               | metres (m)                                     |
| $E$       | Energy of photon/quantum | Joules (J)                                     |

### Applications

- Photoelectric effect explanation (Einstein used  $E = h\nu$ )
- Blackbody radiation curves
- Spectral line emission/absorption

## 2. Dual Nature of Matter (de Broglie)

### DE BROGLIE EQUATION

$$\lambda = h / mv = h / p$$

### KINETIC ENERGY FORM

$$\lambda = h / \sqrt{2mE_k}$$

| Symbol    | Meaning               |
|-----------|-----------------------|
| $\lambda$ | de Broglie wavelength |
| $h$       | Planck's constant     |
| $m$       | Mass of particle      |
| $v$       | Velocity of particle  |
| $p = mv$  | Momentum              |

### Wave-Particle Duality

- Every moving particle has an associated wave
- Larger mass  $\rightarrow$  smaller  $\lambda \rightarrow$  wave nature negligible for macroscopic objects
- Electron has significant wave behavior (confirmed by Davisson-Germer experiment)
- Heisenberg Uncertainty Principle:  $\Delta x \cdot \Delta p \geq h/4\pi$



### 3. Schrödinger Wave Equation

#### TIME-INDEPENDENT SCHRÖDINGER EQUATION

$$\hat{H}\psi = E\psi$$

#### GENERAL (EXPANDED) FORM

$$-(\hbar^2/2m) \nabla^2\psi + V(r)\psi = E\psi$$

#### LAPLACIAN FORM (3D)

$$(\partial^2\psi/\partial x^2) + (\partial^2\psi/\partial y^2) + (\partial^2\psi/\partial z^2) + (8\pi^2m/h^2)(E - V)\psi = 0$$

| Symbol           | Meaning                   | Physical Significance                      |
|------------------|---------------------------|--------------------------------------------|
| $\psi$ (psi)     | Wave function             | Amplitude; contains all quantum info       |
| $\psi^2$         | Probability density       | Probability of finding electron at a point |
| $\hat{H}$        | Hamiltonian operator      | Total energy operator (KE + PE)            |
| $E$              | Total energy              | Eigenvalue of the equation                 |
| $V$              | Potential energy          | Coulombic attraction in atoms              |
| $\hbar = h/2\pi$ | Reduced Planck's constant | $1.055 \times 10^{-34}$ J·s                |

#### ✓ Conditions on $\psi$ (Born Interpretation)

- Must be single-valued, continuous, finite everywhere
- Must be square-integrable:  $\int |\psi|^2 dV = 1$  (normalization)
- $\psi^2 \geq 0$  always; represents **probability density**



### 4. Hydrogen Atom – Energy & Quantum Numbers

#### ENERGY OF NTH LEVEL (H-ATOM)

$$E_n = -13.6 / n^2 \text{ eV}$$

#### RYDBERG FORMULA

$$1/\lambda = R_H(1/n_1^2 - 1/n_2^2)$$

#### BOHR FREQUENCY RULE

$$\Delta E = E_2 - E_1 = h\nu$$

$R_H = 1.097 \times 10^7 \text{ m}^{-1}$  (Rydberg constant)

| Quantum Number | Symbol | Values                | Describes                |
|----------------|--------|-----------------------|--------------------------|
| Principal      | $n$    | 1, 2, 3 ...           | Shell / energy level     |
| Azimuthal      | $l$    | 0 to $n-1$            | Subshell / orbital shape |
| Magnetic       | $m_l$  | $-l \dots 0 \dots +l$ | Orbital orientation      |

Spin

 $m_s$  $+1/2, -1/2$ Electron spin ( $\uparrow$  or  $\downarrow$ )

| l value | Subshell | Orbital shape                | Max electrons |
|---------|----------|------------------------------|---------------|
| 0       | s        | Spherical                    | 2             |
| 1       | p        | Dumbbell (3 orientations)    | 6             |
| 2       | d        | Double dumbbell / cloverleaf | 10            |
| 3       | f        | Complex                      | 14            |

## 5. Molecular Orbital Theory (MOT)

### BOND ORDER FORMULA

$$\text{B.O.} = (N_b - N_a) / 2$$

### LCAO Principle

Molecular Orbitals are formed by **Linear Combination of Atomic Orbitals**

$$\Psi_{\text{bonding}} = \Psi_A + \Psi_B \quad \Psi_{\text{antibonding}} = \Psi_A - \Psi_B$$

| B.O. | Stability       | Example                         |
|------|-----------------|---------------------------------|
| 0    | Does NOT exist  | He <sub>2</sub>                 |
| 1    | Single bond     | H <sub>2</sub> , F <sub>2</sub> |
| 2    | Double bond     | O <sub>2</sub>                  |
| 3    | Triple bond     | N <sub>2</sub> , CO             |
| > 0  | Stable molecule | —                               |

| Orbital Type   | Symbol            | Energy         | Electron density    | Effect on bond   |
|----------------|-------------------|----------------|---------------------|------------------|
| Bonding MO     | $\sigma, \pi$     | Lower than AO  | High between nuclei | Strengthens bond |
| Antibonding MO | $\sigma^*, \pi^*$ | Higher than AO | Node between nuclei | Weakens bond     |
| Non-bonding MO | n                 | Same as AO     | No change           | No effect        |

## 6. Diatomic Molecule MO Configurations

Energy Level Order (for O<sub>2</sub>, F<sub>2</sub>, Ne<sub>2</sub> — Z > 7):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p < \pi 2p = \pi 2p < \pi^* 2p = \pi^* 2p < \sigma^* 2p$$

Energy Level Order (for Li<sub>2</sub> to N<sub>2</sub> — Z ≤ 7):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p = \pi 2p < \sigma 2p < \pi^* 2p = \pi^* 2p < \sigma^* 2p$$

| Molecule        | Total e <sup>-</sup> | N <sub>b</sub> | N <sub>a</sub> | Bond Order | Bond Type      | Magnetic            |
|-----------------|----------------------|----------------|----------------|------------|----------------|---------------------|
| H <sub>2</sub>  | 2                    | 2              | 0              | 1          | Single         | Diamagnetic         |
| He <sub>2</sub> | 4                    | 2              | 2              | 0          | Does not exist | —                   |
| N <sub>2</sub>  | 14                   | 10             | 4              | 3          | Triple         | Diamagnetic         |
| O <sub>2</sub>  | 16                   | 10             | 6              | 2          | Double         | <b>Paramagnetic</b> |
| F <sub>2</sub>  | 18                   | 10             | 8              | 1          | Single         | Diamagnetic         |
| CO              | 14                   | 10             | 4              | 3          | Triple         | Diamagnetic         |
| NO              | 15                   | 10             | 5              | 2.5        | 2.5            | Paramagnetic        |
| NO <sup>+</sup> | 14                   | 10             | 4              | 3          | Triple         | Diamagnetic         |

### Magnetic Behavior Rule

- Paramagnetic** → Unpaired electrons in MOs (attracted to magnetic field)

- **Diamagnetic** → All electrons paired (repelled by magnetic field)
- O<sub>2</sub> is paramagnetic because 2 electrons in degenerate  $\pi^*2p$  orbitals are unpaired (Hund's rule)

## 7. $\pi$ -Molecular Orbitals

### 1,3-Butadiene ( $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ )

- 4 carbon atoms  $\rightarrow$  4  $\pi$ -MOs
- $\psi_1, \psi_2$  = Bonding MOs (occupied)
- $\psi_3^*, \psi_4^*$  = Antibonding MOs (vacant)
- HOMO =  $\psi_2$  | LUMO =  $\psi_3^*$
- Nodes:  $\psi_1=0, \psi_2=1, \psi_3^*=2, \psi_4^*=3$
- Delocalization over 4 carbons  $\rightarrow$  stability

### Benzene ( $\text{C}_6\text{H}_6$ )

- 6 carbon atoms  $\rightarrow$  6  $\pi$ -MOs
- 3 Bonding + 3 Antibonding
- $\psi_1$  (lowest),  $\psi_2, \psi_3$  = bonding (filled)
- $\psi_4^*, \psi_5^*, \psi_6^*$  = antibonding (empty)
- 6  $\pi$ -electrons fully delocalized in ring
- Extra stability = **Resonance energy**
- Aromaticity:  $4n+2$   $\pi$ -electrons ( $n=1 \rightarrow 6$ )

| Concept           | Butadiene                  | Benzene                              |
|-------------------|----------------------------|--------------------------------------|
| No. of p-orbitals | 4                          | 6                                    |
| No. of MOs        | 4                          | 6                                    |
| Bonding MOs       | 2 ( $\psi_1, \psi_2$ )     | 3 ( $\psi_1, \psi_2, \psi_3$ )       |
| Antibonding MOs   | 2 ( $\psi_3^*, \psi_4^*$ ) | 3 ( $\psi_4^*, \psi_5^*, \psi_6^*$ ) |
| Delocalization    | Over 4 C atoms             | Over entire ring                     |
| Special property  | Conjugation                | Aromaticity                          |

### Key Concepts: Conjugation & Delocalization

- **Conjugation:** Alternating single and double bonds allowing  $\pi$ -electron overlap
- **Delocalization:**  $\pi$ -electrons spread over multiple atoms  $\rightarrow$  greater stability
- **HOMO** = Highest Occupied Molecular Orbital (donor role)
- **LUMO** = Lowest Unoccupied Molecular Orbital (acceptor role)
- Number of nodes increases with energy:  $\psi_1$  has 0 nodes (most stable)

## 8. Electrochemistry – Fundamentals

### EMF OF CELL

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

### STANDARD EMF

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

### NERNST EQUATION (298 K)

$$E = E^{\circ} - (0.0591/n) \log Q$$

### NERNST – GENERAL FORM

$$E = E^{\circ} - (RT/nF) \ln Q$$

### GIBBS ENERGY & EMF

$$\Delta G^{\circ} = -nFE^{\circ}$$

### EQUILIBRIUM & EMF

$$\log K_{\text{eq}} = nE^{\circ}/0.0591$$

| Symbol      | Meaning                      | Value / Unit                                    |
|-------------|------------------------------|-------------------------------------------------|
| E           | Cell potential (EMF)         | Volts (V)                                       |
| $E^{\circ}$ | Standard electrode potential | Volts (V) at 298 K, 1 M, 1 atm                  |
| n           | No. of electrons transferred | mol (dimensionless)                             |
| F           | Faraday's constant           | 96485 C mol <sup>-1</sup> $\approx$ 96500 C/mol |
| R           | Gas constant                 | 8.314 J mol <sup>-1</sup> K <sup>-1</sup>       |
| T           | Temperature                  | Kelvin (K)                                      |
| Q           | Reaction quotient            | [products]/[reactants]                          |

## Galvanic vs Electrolytic Cell

| Feature           | Galvanic (Voltaic) Cell           | Electrolytic Cell                   |
|-------------------|-----------------------------------|-------------------------------------|
| Energy            | Chemical $\rightarrow$ Electrical | Electrical $\rightarrow$ Chemical   |
| Spontaneous?      | Yes ( $\Delta G < 0$ )            | No ( $\Delta G > 0$ )               |
| $E_{\text{cell}}$ | Positive ( $> 0$ )                | Negative ( $< 0$ ), external needed |
| Anode             | Negative (oxidation)              | Positive (oxidation)                |
| Cathode           | Positive (reduction)              | Negative (reduction)                |
| Examples          | Daniel cell, Batteries            | Electroplating, Electrolysis        |

### Memory Tricks

- AN OX = ANode  $\rightarrow$  OXidation | RED CAT = REDuction  $\rightarrow$  CATHode
- In galvanic cell: **Anode is -ve, Cathode is +ve**
- In electrolytic cell: **Anode is +ve, Cathode is -ve**

- Salt bridge maintains electrical neutrality

## ⚡ 9. Electrochemical Applications

### 🔋 Batteries

- **Primary:** Non-rechargeable (dry cell, alkaline)
- **Secondary:** Rechargeable (Lead-acid, Li-ion)
- **Lead-Acid:**  $E^\circ \approx 2V/\text{cell}$ ; 12V = 6 cells
- **Fuel Cell:**  $H_2 + O_2 \rightarrow H_2O + \text{electricity}$

### 🔧 Electroplating

- Object to plate = **Cathode**
- Plating metal = **Anode**
- Electrolyte = salt of plating metal
- Mass deposited:  **$m = ZIt$**
- $Z$  = Electrochemical equivalent

### 🦋 Corrosion

- Electrochemical process (galvanic)
- Anode:  $Fe \rightarrow Fe^{2+} + 2e^-$
- Cathode:  $O_2 + H_2O + 2e^- \rightarrow OH^-$
- Rust =  $Fe_2O_3 \cdot xH_2O$
- Prevention: Galvanizing (Zn coating)
- Cathodic protection method

## Faraday's Laws of Electrolysis

### 1ST LAW – MASS DEPOSITED

$$m = Z \times I \times t$$

### 2ND LAW – EQUIVALENT WEIGHTS

$$m_1/m_2 = E_1/E_2$$

| Symbol | Meaning                                | Unit    |
|--------|----------------------------------------|---------|
| $m$    | Mass deposited                         | grams   |
| $Z$    | Electrochemical equivalent = $E/96500$ | g/C     |
| $I$    | Current                                | Amperes |
| $t$    | Time                                   | seconds |
| $E$    | Equivalent weight                      | g/mol   |

### Conductance Formulas

#### SPECIFIC CONDUCTANCE

$$\kappa = l / (R \times A)$$

#### MOLAR CONDUCTANCE

$$\Lambda_m = \kappa \times 1000 / M$$

- Kohlrausch's Law:  $\Lambda_m^\circ = \lambda_+^\circ + \lambda_-^\circ$

## 10. Exam Flashcards (25 Q&A)

Q: Planck's energy equation?

A:  $E = h\nu$

Q: de Broglie equation?

A:  $\lambda = h / mv = h / p$

Q: Nernst equation at 298 K?

A:  $E = E^\circ - (0.0591/n) \log Q$

Q: Schrödinger equation (short)?

A:  $\hat{H}\psi = E\psi$

Q: Is  $O_2$  paramagnetic? Why?

A: Yes – 2 unpaired  $e^-$  in  $\pi^*2p$  orbitals

Q: HOMO and LUMO?

A: Highest Occupied / Lowest Unoccupied MO

Q: What is Faraday's constant?

A:  $F = 96500 \text{ C/mol}$

Q: Anode process in galvanic cell?

A: Oxidation (negative electrode)

Q: Planck's constant value?

A:  $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

Q: Hund's rule in MO theory?

A: Fill degenerate orbitals singly first

Q: Aromaticity rule (Hückel)?

A:  $4n+2$   $\pi$ -electrons ( $n = 0, 1, 2, \dots$ )

Q: Specific conductance formula?

A:  $\kappa = l / (R \times A)$

Q: Rydberg constant value?

Q: What does  $\psi^2$  represent?

A: Probability density of finding electron

Q: Bond order formula?

A:  $B.O. = (N_b - N_a) / 2$

Q: Gibbs energy and EMF?

A:  $\Delta G^\circ = -nFE^\circ$

Q: Energy of H-atom nth level?

A:  $E_n = -13.6 / n^2 \text{ eV}$

Q: B.O. of  $N_2$ ?

A: 3 (triple bond, diamagnetic)

Q: Faraday's 1st law?

A:  $m = ZIt$  (mass  $\propto$  charge)

Q: LCAO stands for?

A: Linear Combination of Atomic Orbitals

Q: Cathode process in electrolytic?

A: Reduction (negative electrode)

Q: Normalization condition?

A:  $\int |\psi|^2 dV = 1$

Q: B.O. of  $He_2$ ?

A: 0 (does not exist)

Q: Heisenberg uncertainty?

A:  $\Delta x \cdot \Delta p \geq h/4\pi$

Q: What is  $\psi$  (wave function)?

A: Mathematical function containing all quantum info

$$\text{A: } R_H = 1.097 \times 10^7 \text{ m}^{-1}$$

## 11. Exam Tips & Common Mistakes

### Quantum Mechanics Pitfalls

- ✗ **Confusing  $\psi$  with  $\psi^2$ :**  $\psi$  is the wave function (amplitude).  $\psi^2$  is probability density. Only  $\psi^2$  has physical meaning.  $\psi$  can be negative!
- ✗ **Wrong Quantum Number ranges:**  $l$  goes from 0 to  $n-1$  (NOT 1 to  $n$ ).  $m_l$  goes from  $-l$  to  $+l$  including 0. Total =  $2l+1$  values.
- ✗ **de Broglie for massive objects:** For large mass,  $\lambda$  is so tiny it's negligible. Only electrons/protons show wave nature experimentally.

### MO Theory Pitfalls

- ✗ **Forgetting antibonding electrons:** Every antibonding electron reduces bond order. Don't count only bonding electrons in the formula!
- ✗ **Wrong MO energy order:** For  $N_2$  ( $Z \leq 7$ ):  $\pi 2p$  comes BEFORE  $\sigma 2p$ . For  $O_2$  ( $Z > 7$ ):  $\sigma 2p$  comes BEFORE  $\pi 2p$ . Use the correct order!
- ✗ **Ignoring Hund's rule in MOs:** Degenerate  $\pi^* 2p$  orbitals in  $O_2$  get one electron each first  $\rightarrow$  paramagnetic. This is why  $O_2$  has B.O. = 2 AND is paramagnetic.
- ✗ **Wrong antibonding notation:** Antibonding MOs are denoted with  $*$  (sigma-star  $\sigma^*$ , pi-star  $\pi^*$ ). Don't write them without the asterisk!

### Electrochemistry Pitfalls

- ✗ **Nernst sign error:**  $E = E^\circ \text{ MINUS } (0.0591/n) \log Q$ . The sign is NEGATIVE. A common mistake is writing  $E = E^\circ + \dots$  which is wrong.
- ✗ **Q vs K in Nernst equation:**  $Q$  is the reaction quotient (any conditions). When  $Q = K$ ,  $E = 0$  (equilibrium). Don't substitute  $K$  in Nernst unless at equilibrium.
- ✗ **Galvanic vs Electrolytic anode charge:** Galvanic  $\rightarrow$  anode is  $-ve$ . Electrolytic  $\rightarrow$  anode is  $+ve$ . These are OPPOSITE! Don't mix them up.
- ✗  **$\Delta G^\circ = -nFE^\circ$ :** Negative sign is crucial. If  $E^\circ$  is positive (spontaneous),  $\Delta G^\circ$  is negative. If you forget the sign, your  $\Delta G^\circ$  will have wrong sign!

### Golden Rules for Exams

- Always write units with constants ( $h$  in  $J \cdot s$ ,  $F$  in  $C/mol$ ,  $R$  in  $J/mol \cdot K$ )
- Bond order must be calculated using BOTH bonding and antibonding electrons
- For magnetic behavior:  $>0$  unpaired  $e^-$  = paramagnetic; all paired = diamagnetic
- Kohlrausch's law applies at INFINITE dilution for weak electrolytes

- $\psi$  must be normalized, single-valued, continuous, and finite



## 12. Quick Revision Master Sheet – All Formulas



### QUANTUM THEORY

Planck's Energy

$$E = h\nu = hc/\lambda$$

de Broglie Wavelength

$$\lambda = h/mv = h/p$$

Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq h/4\pi$$

Planck's Constant

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

de Broglie (Energy form)

$$\lambda = h/\sqrt{2mEk}$$

Reduced Planck

$$\hbar = h/2\pi = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}$$



### SCHRÖDINGER EQUATION

Compact Form

$$\hat{H}\psi = E\psi$$

Normalization

$$\int |\psi|^2 dV = 1$$

3D Full Form

$$\nabla^2\psi + (8\pi^2m/h^2)(E-V)\psi = 0$$

Probability Density

$$P = \psi^2$$



### HYDROGEN ATOM

Energy (nth level)

$$E_n = -13.6/n^2 \text{ eV}$$

Rydberg Constant

$$RH = 1.097 \times 10^7 \text{ m}^{-1}$$

Rydberg Formula

$$1/\lambda = RH(1/n_1^2 - 1/n_2^2)$$

Bohr Frequency

$$\Delta E = E_2 - E_1 = h\nu$$



### MOLECULAR ORBITAL THEORY

Bond Order

$$B.O. = (N_b - N_a)/2$$

Antibonding MO

$$\psi^* = \psi_A - \psi_B$$

N<sub>2</sub> Bond Order

$$B.O. = 3 \text{ (diamagnetic)}$$

Bonding MO

$$\psi_b = \psi_A + \psi_B$$

O<sub>2</sub> Bond Order

$$B.O. = 2 \text{ (paramagnetic)}$$

CO Bond Order

$$B.O. = 3 \text{ (diamagnetic)}$$



### ELECTROCHEMISTRY

Cell EMF

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

Nernst (General)

$$E = E^\circ - (RT/nF) \ln Q$$

Equilibrium & EMF

$$\log K_{\text{eq}} = nE^\circ/0.0591$$

Faraday's 1st Law

$$m = ZIt$$

Nernst (298 K)

$$E = E^\circ - (0.0591/n) \log Q$$

Gibbs & EMF

$$\Delta G^\circ = -nFE^\circ$$

Faraday Constant

$$F = 96500 \text{ C/mol}$$

Faraday's 2nd Law

$$m_1/m_2 = E_1/E_2$$

Specific Conductance

$$\kappa = \ell / (R \cdot A)$$

Molar Conductance

$$\Lambda_m = \kappa \times 1000 / M$$

Kohlrausch's Law

$$\Lambda^{\circ}_m = \lambda^{\circ}(+) + \lambda^{\circ}(-)$$

Electroch. Equivalent

$$Z = E / 96500$$



**You've got this! All the best for your exam! ⚡**

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